The stationary numbers are given by $-\frac{2x}{(1+x^2)^2} = 0$ that is $x = 0$.

Hence the intervals of monotonicity are $(-\infty, 0)$ and $(0, \infty)$.

On the interval $(-\infty, 0)$ the function strictly increases because $f'(x) > 0$ in that interval.

The function $f(x)$ strictly decreases in the interval $(0, \infty)$ because $f'(x) < 0$ in that interval. Since $f'(x)$ changes from positive to negative when passing through $x = 0$, the first derivative test tells us there is local maximum at $x = 0$ and the local maximum value is $f(0) = 1$.

Example 7.56

Find the intervals of monotonicity and local extrema of the function $f(x) = \frac{x}{1+x^2}$.

Solution

The given function is defined and differentiable at all $x \in (-\infty, \infty)$, As

$$f(x) = \frac{x}{1+x^2},$$

$$f'(x) = \frac{1-x^2}{(1+x^2)^2}.$$

The stationary numbers are given by $1 - x^2 = 0$ that is $x = \pm 1$.

Hence the intervals of monotonicity are $(-\infty, -1), (-1, 1)$ and $(1, \infty)$.

Interval	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
Sign of $f'(x)$	$-$	$+$	$-$
Monotonicity	strictly decreasing	strictly increasing	strictly decreasing

Table 7.6

Therefore, $f(x)$ strictly increasing on $(-\infty, -1)$ and $(1, \infty)$, strictly decreasing on $(-1, 1)$.

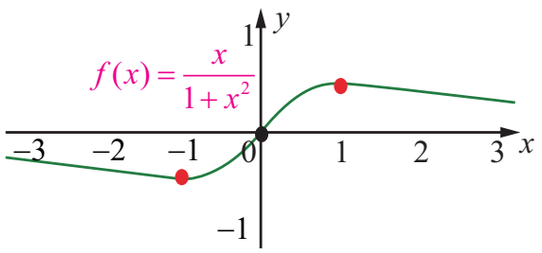


Fig.7.22

Since $f'(x)$ changes from negative to positive when passing through $x = -1$, the first derivative test tells us there is a local minimum at $x = -1$ and the local minimum value is $f(-1) = -\frac{1}{2}$. As $f'(x)$ changes from positive to negative when passing through $x = 1$, the first derivative test tells us there is a local maximum at $x = 1$ and the local maximum value is $f(1) = \frac{1}{2}$.

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